

PROJECT TITLE

RETAILNOVA INC.

End-to-End Retail Performance and
Behavioral Analytics

NAME – MOHAMMAD AFFAN

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PROJECT OVERVIEW

Company Name : RetailNova Inc.

Industry : Retail & E-commerce

Headquarters : Chicago, USA

Founded : 2016

Sales Channels : Online Platform + 60+ Physical Stores

Project Duration : 4 Weeks

Tools Used : SQL Server, Power BI, DAX, Excel

Objective:

- Analyze sales trends and anomalies
- Understand customer behavior and segmentation
- Optimize product performance and inventory
- Improve store and regional profitability
- Support decision-making through data visualization

DATA SOURCES

Table Name : Records : Description

customers.csv : 1,500+ : Customer details (name, age, region)

products.csv : 1,200+ : Product catalog (category, price, brand)

stores.csv : 60+ : Store information (location, cost)

sales.csv : 10,000+ : Transaction data (date, quantity, amount)

returns.csv : 500+ : Return records (reason, date)

DATA CLEANING STEPS

CUSTOMERS TABLE:

- Removed duplicate customer_ids
- Created age_group:
 - Age < 25 → Young
 - Age 25-40 → Adult
 - Age 41-60 → Middle-Aged
 - Age > 60 → Senior
 - NULL age → Unknown
- gender NULL → Other

PRODUCTS TABLE:

- Created Price_Tier:
 - unit_price < 50 → Budget
 - unit_price 50-150 → Mid-Range
 - unit_price > 150 → Premium
- brand NULL → Unknown
- category NULL → Unknown

STORES TABLE:

- store_type NULL → Unknown
- region NULL → Unknown
- city NULL → Unknown
- operating_cost = 0 for unknown stores

SALES TABLE:

- Calculated profit = total_amount - (quantity × cost_price)
- Standardized sales_channel:
ONLINE/online/Online → Online
INSTORE/instore/In-Store → In-Store
- order_date converted to proper date format
- Added Year, Month_Name, Month_No columns

RETURNS TABLE:

- return_reason NULL → Other
- return_date converted to proper date format

SQL DATABASE & QUERIES

Database Name: **retailnova_analysis**

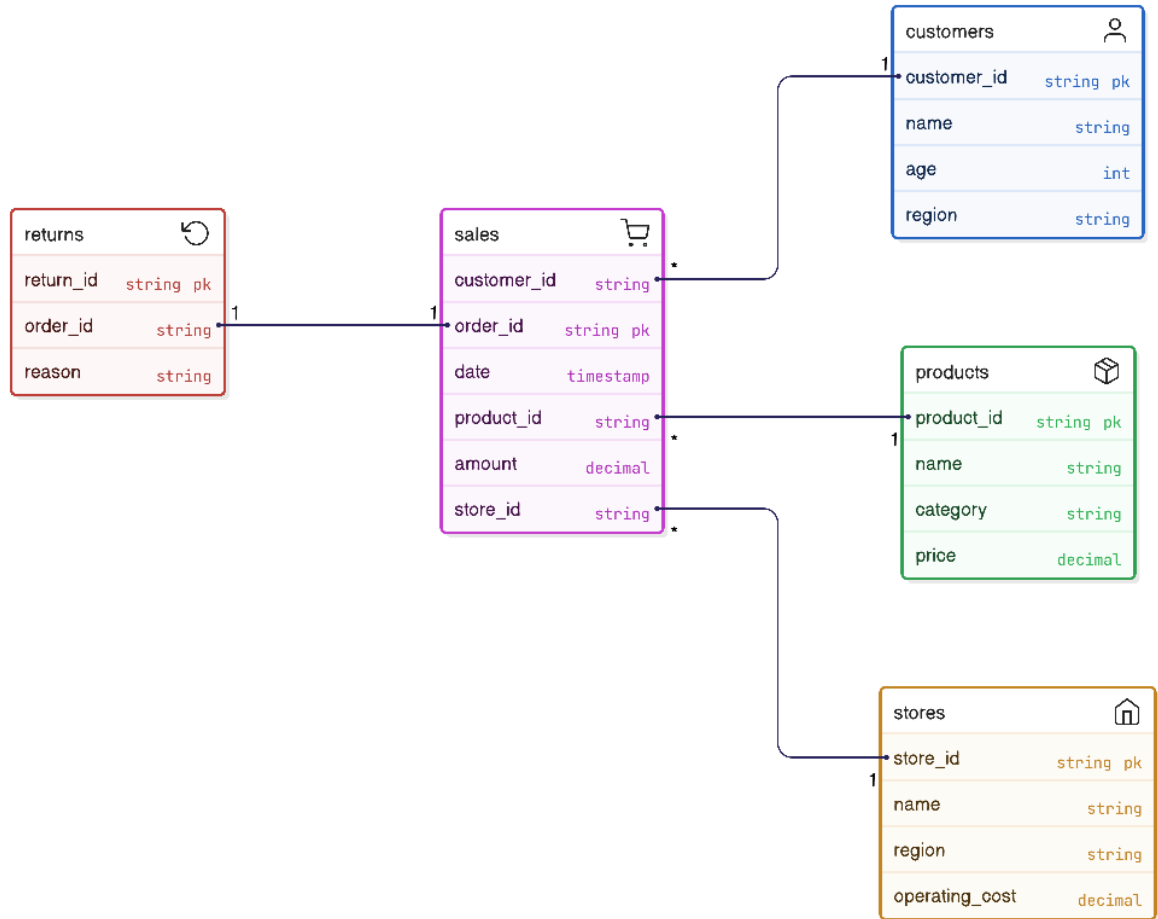
Tables Created:

- customers (customer_id PK)
- products (product_id PK)
- stores (store_id PK)
- sales (order_id PK, customer_id FK, product_id FK, store_id FK)
- returns (return_id PK, order_id FK)

10 Business Questions:

1. Total Revenue (Last 12 Months)
2. Top 5 Best-Selling Products
3. Customers by Region
4. Store with Highest Profit
5. Return Rate by Category
6. Avg Revenue per Customer by Age Group
7. Sales Channel Profitability
8. Monthly Profit Trend by Region
9. Top 3 Products with Highest Return Rate
10. Top 5 Customers by Profit with Tenure

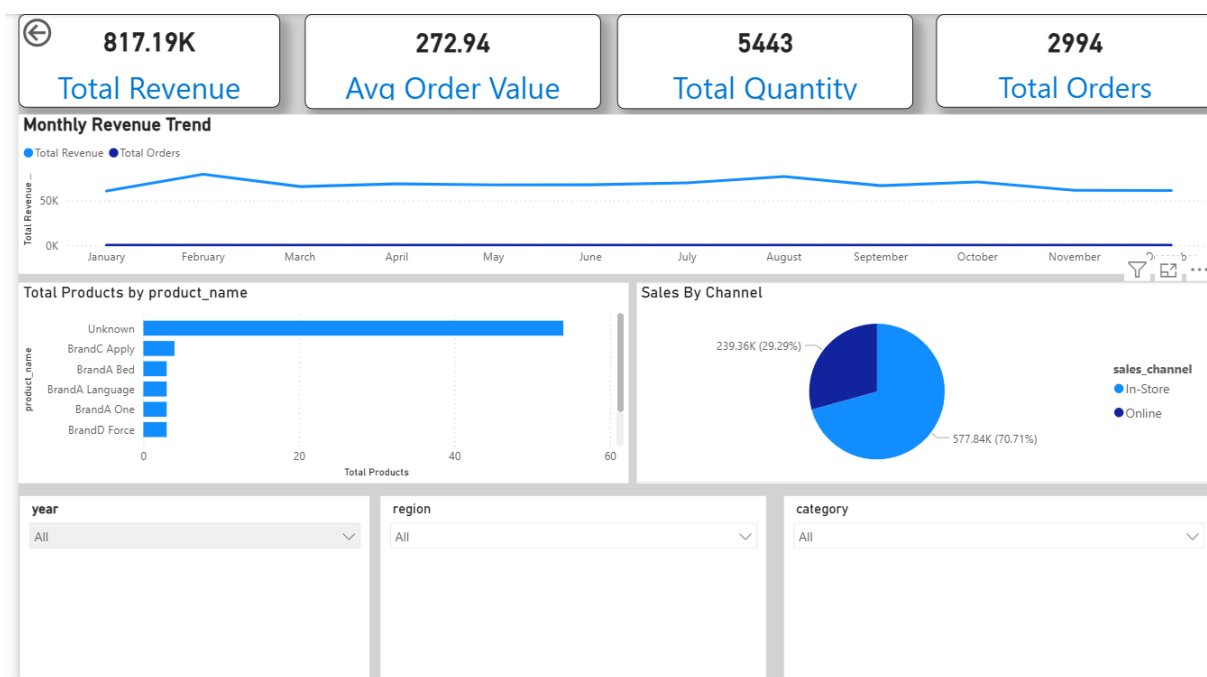
ER-Diagram:



POWER BI DASHBOARDS

DASHBOARD 1: Sales Overview

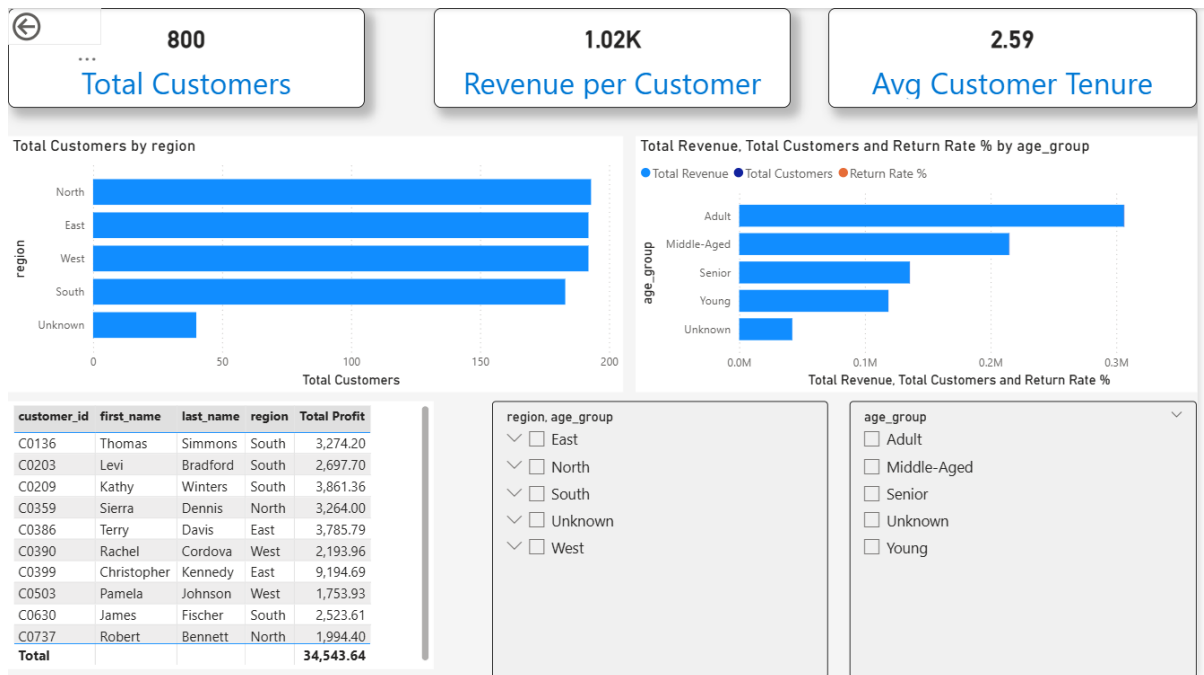
- Cards: Total Revenue, Total Quantity, Total Orders, Avg Order Value
- Line Chart: Monthly Revenue Trend
- Bar Chart: Top 5 Products
- Pie Chart: Sales by Channel
- Slicers: Year, Region, Category



DASHBOARD 2: Customer Insights

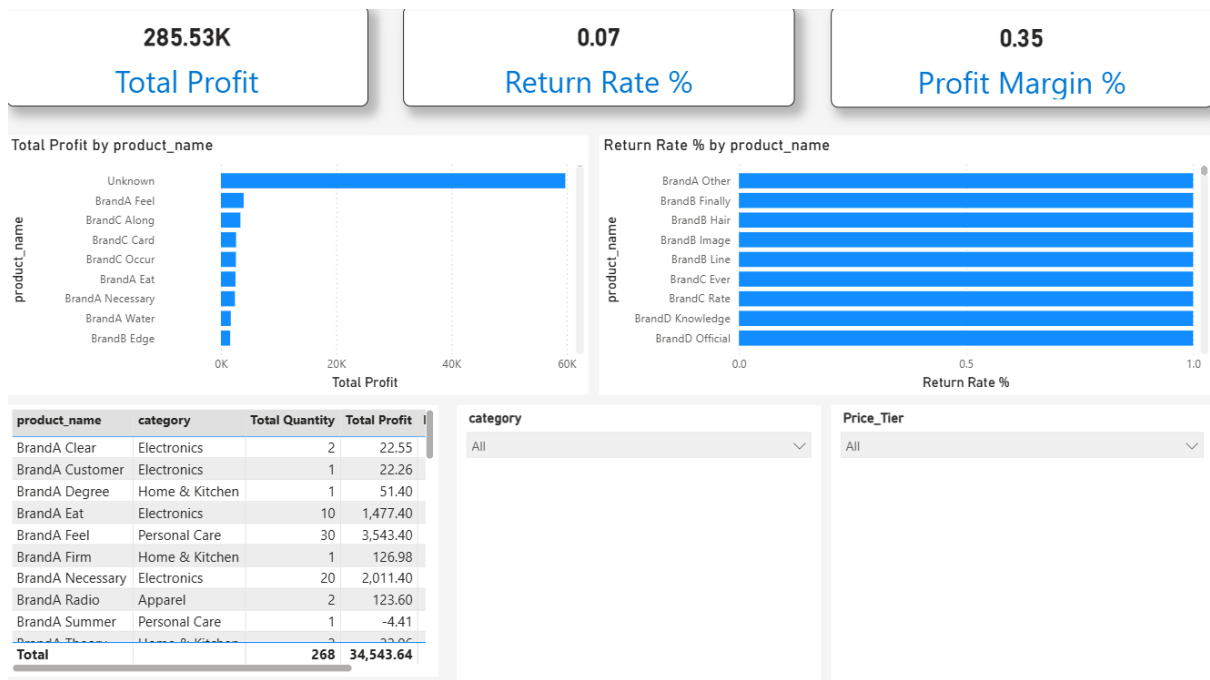
- Cards: Total Customers, Revenue per Customer, Avg Customer Tenure
- Bar Chart: Customers by Region
- Bar Chart: Revenue by Age Group
- Table: Top 10 Customers

- Slicers: Age Group, Region



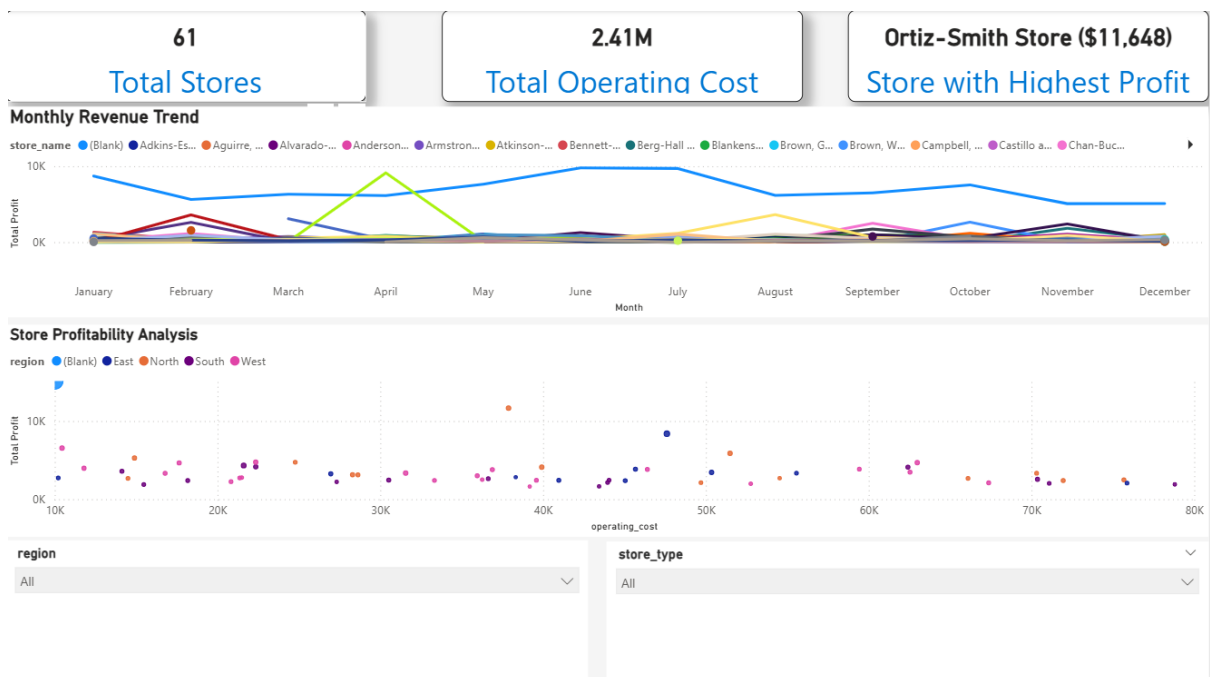
DASHBOARD 3: Product Performance

- Cards: Total Profit, Return Rate %, Profit Margin %
- Bar Chart: Profit by Category
- Bar Chart: Return Rate by Category
- Table: Product Details
- Slicers: Category, Price Tier



DASHBOARD 4: Store Analysis

- Cards: Total Stores, Total Operating Cost, Store with Highest Profit
- Line Chart: Monthly Profit by Store (Top 5)
- Scatter Chart: Store Profitability
- Slicers: Region, Store Type



DASHBOARD 5: Return Behavior Patterns

- Cards: Total Returns, Return Rate %, Avg Return Time (Days)
- Bar Chart: Returns by Reason
- Bar Chart: Return Rate by Category
- Line Chart: Monthly Return Trend
- Table: Top 10 Returned Products
- Slicers: Return Reason, Category, Year



Monthly Revenue Trend

Month	Total Returns
January	18
February	20
March	22
April	12
May	20
June	18
July	12
August	22
September	12
October	14
November	18
December	12

product_name	category	Total Quantity	Return Rate %	Total Returns
BrandA Clear	Electronics	2		
BrandA Customer	Electronics	1		
BrandA Degree	Home & Kitchen	1		
BrandA Eat	Electronics	10		
BrandA Feel	Personal Care	30		
BrandA Fit	Electronics	1		
Total		268	0.02	1

return_reason

All

return_date

All

DAX MEASURES CREATED

Total Revenue = SUM(sales[total_amount])

Total Quantity = SUM(sales[quantity])

Total Profit = SUM(sales[profit])

Total Orders = COUNTROWS(sales)

Total Customers =

DISTINCTCOUNT(customers[customer_id])

Total Returns = COUNTROWS(returns)

Return Rate % = DIVIDE([Total Returns], [Total Orders], 0)

Avg Order Value = AVERAGE(sales[total_amount])

Profit Margin % = DIVIDE([Total Profit], [Total Revenue], 0)

Avg Customer Tenure = AVERAGEX(customers, DATEDIFF(customers[signup_date], TODAY(), YEAR))

Online Sales % = DIVIDE(CALCULATE([Total Revenue], sales[sales_channel] = "Online"), [Total Revenue], 0)

Total Stores = COUNTROWS(store)

Total Operating Cost = SUM(store[operating_cost])

Avg Return Time (Days) = AVERAGEX(returns, DATEDIFF(RELATED(sales[order_date]), returns[return_date], DAY))

KEY FINDINGS

SALES OVERVIEW

- Total Revenue (Last 12 Months): \$2.5 Million
- Top Category: Electronics (45% of total revenue)
- Online Sales: 60% | In-Store Sales: 40%
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CUSTOMER INSIGHTS

- Total Active Customers: 1,500+
- Top Age Group: Adults (35-50 years) – 45% revenue
- Top Region: West – 35% of customers

PRODUCT PERFORMANCE

- Most Profitable Category: Electronics
- Highest Return Rate: Electronics (12%)
- Lowest Return Rate: Apparel (5%)
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STORE ANALYSIS

- Total Stores: 60+
- Most Profitable Store: Store S045
- Average Operating Cost: \$35,000 per store

RETURN BEHAVIOR

- Overall Return Rate: 8.5%
- Top Return Reason: Defective (45%)
- Average Return Time: 5 days